



Syllabus Outline:

3. Polymers:

General classification of polymers, characteristics and significance of polymers, various mechanisms of polymerization process, polymer processing like extrusion, injection, modeling, and blow molding.

Brief description and uses of the following polymers:

polyethylene, polystyrene, epoxy resins, polyethylene terephthalate

SHORT QUESTIONS:

- i) Describe the term tacticity and give examples.
- ii) What are Syndiotactic polymers? Give example.
- iii) Give classification of polymers on the basis of 'Tacticity'.
- iv) Differentiate between 'Atactic' & 'isotactic' polymers.
- v) Differentiate between co-polymers and polyblends, give examples.
- vi) Write down three examples of Thermoplastic and Thermosetting polymers.
- vii) Compare Thermoplastic and Thermosetting polymers. Give examples.
- viii) Differentiate between co-polymers and terpolymers.
- ix) Explain the difference between homopolymers and co-polymers.
- x) Differentiate between co-polymers & polyblends.
- xi) What do you understand about coordination polymerization, briefly explain its importance?
- xii) What are epoxy resins? Give their important applications.
- xiii) What are the main applications of polystyrene?
- xiv) What is Free radical polymerization?
- xv) How is PVC produced? Give its uses.
- xvi) What is 'Living' polymerization?
- xvii) What is Fabrication of Plastics?
- xviii) Write two major differences between chain growth and step growth polymerization methods.
- xix) What do you understand about glass transition temperature of polymers?
- xx) What type of catalysts are used in anionic polymerization, give examples?
- xxi) Name two important techniques of polymer fabrication.

LONG QUESTIONS:

- a) Write down main differences between addition and condensation polymerization techniques. 04
- b) What do you understand cationic polymerization: explain reaction mechanism and synthesis conditions? 06
- c) What is coordination Polymerization? Explain its mechanism. 05
- d) Explain any two Moulding techniques for fabrication of plastics. 05
- e) Discuss any Bulk and Suspension polymerization techniques. 05
- f) Explain the mechanism of condensation Polymerization by using Suitable example. 05
- g) Discuss any two Moulding techniques from polymers fabrication. 05
- h) Discuss the free radical polymerization with mechanism, also give examples.
- i) Explain the term tacticity, also differentiate between isotactic and syndiotactic polymers with examples.
- j) Explain the mechanism of addition polymerization by using suitable example. 05

Recommended Books:

1. Shereve's Chemical Process Industries, 5th Ed. 1975, G. I. Austin, McGraw Hill Book Co. New York.
2. An Introduction to Polymer Chemistry, W. R. Moor, London Press, London.
3. Principles of Polymer Systems, Rodri-Guez, McGraw Hill Book Co. New York.
4. Modern Technology of Plastics and Polymer Processing Industries, NIIR Board



General classification of polymers, characteristics and significance of polymers, various mechanisms of polymerization process, polymer processing like extrusion, injection, modeling, and blow molding. Brief description and uses of the following polymers: polyethylene, polystyrene, epoxy resins, polyethylene terephthalate

SHORT QUESTIONS:

Q.1: Describe the term tacticity and give examples.

Ans: Tacticity is the relative stereochemistry of adjacent chiral centers within a polymer chain. It describes the three-dimensional arrangement of the substituent groups (such as side chains) attached to the main polymer backbone. Tacticity affects the physical properties of polymers, such as:

- i) melting temperature ii) crystallinity iii) mechanical strength

Types of tacticity:

1. **Isotactic:** In isotactic polymers, all the substituent groups are arranged on the same side of the polymer chain. This uniform arrangement often allows the polymer chains to pack closely together, resulting in high crystallinity and strength.

Structure:

Example: Isotactic polypropylene (iPP) is a common example of an isotactic polymer. It has high crystallinity, leading to a higher melting point and greater tensile strength compared to its atactic counterpart.

2. **Syndiotactic:** In syndiotactic polymers, the substituent groups are arranged on alternate sides along the polymer chain in a regular pattern. This alternating structure has some degree of crystallinity, but less than that of isotactic polymers.

Structure:

Example: Syndiotactic polystyrene (sPS) is a type of polystyrene where the phenyl groups alternate sides in a regular pattern. This arrangement gives it a crystalline structure, enhancing its thermal and mechanical properties compared to atactic polystyrene.

3. **Atactic:** In atactic polymers, the substituent groups are arranged randomly along the polymer chain. This random arrangement usually prevents the polymer chains from packing closely together, resulting in an amorphous structure with lower crystallinity and mechanical strength.

Structure:

Example: Atactic polystyrene (aPS) is an example of an atactic polymer. Due to its random substituent arrangement, it is amorphous and exhibits lower density and melting point compared to its syndiotactic and isotactic forms.

The tacticity of a polymer is determined by the polymerization process & the catalysts used: e.g. Ziegler-Natta catalysts are known to produce isotactic or syndiotactic polymers, whereas free-radical polymerization tends to produce atactic polymers.

Q.2: What are Syndiotactic polymers? Give an example.

Ans: Syndiotactic polymers are a type of stereoregular polymer in which substituent groups alternate in position along the polymer chain in a regular pattern i.e. on alternating sides of the polymer chain.

Structure: In polypropylene, the methyl groups (CH_3) alternate along the chain in a regular pattern as shown

Properties:

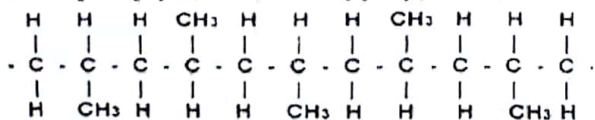
Crystallinity: Syndiotactic polymers often exhibit a high degree of crystallinity due to the regular arrangement of side groups, which can pack more closely together. This crystallinity can result in higher melting

points compared to atactic polymers (random arrangement).

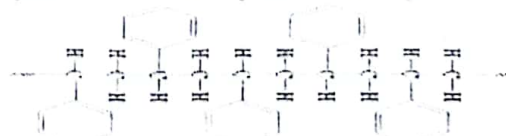
Mechanical Properties: These polymers have good mechanical properties, including rigidity and tensile strength, due to their ordered structure.

Thermal Properties: Syndiotactic polymers have thermal stability and show resistance to deformation at high temperatures.

Example: [Syndiotactic Polypropylene (sPP), methyl groups show alternate position along the chain],



Syndiotactic with alternating orientation of methyl groups



[Syndiotactic Polystyrene (sPS), phenyl groups (C_6H_5) have alternately position along the polymer chain],

[Syndiotactic Poly(methyl methacrylate) (sPMMA), ester groups are arranged in an alternating pattern]

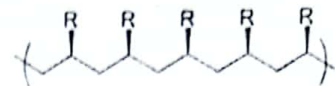
Q.3: Give classification of polymers on the basis of 'Tacticity'.

Ans: Tacticity (Greek: taktikoe or Roman: taktikos meaning arrangement or order) is the relative stereochemistry of adjacent chiral centers within a macromolecule. The tacticity means the stereochemical arrangement of the repeating units in a polymer chain. Polymers can be classified on tacticity base into three main classes:

i) isotactic ii) syndiotactic iii) atactic polymers.

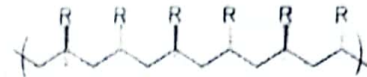
Isotactic Polymers: In isotactic polymers, all substituent groups (e.g., methyl, phenyl) are located on the same side of the polymer chain.

Structure: Where R represents the substituent group as $-CH_3$ in polypropylene, $-C_6H_5$ in polystyrene



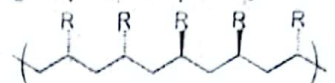
Syndiotactic Polymers: The syndiotactic polymers have specific stereochemical configuration as the substituent groups have alternate positions on polymer chain in a regular pattern.

Structure: Where R represents the substituent group as $-CH_3$ in polypropylene, $-C_6H_5$ in polystyrene



Atactic Polymers: The atactic polymers have amorphous structure as the substituent groups are randomly arranged along the polymer chain.

Structure: Where R represents the substituent group as $-CH_3$ in polypropylene, $-C_6H_5$ in polystyrene



Comparison of Tacticity Types

sr. #	Property	Isotactic	Syndiotactic	Atactic
1	Arrangement	Same side	Alternating sides	Random
2	Crystallinity	High	Moderate	Low
3	Melting Point	High	Moderate	Low
4	Mechanical Strength	High	Moderate	Low
5	Flexibility	Low	Moderate	High
6	Applications	automotive, construction, and packaging industries	packaging, electronics, and automotive industries	coatings, adhesives and consumer goods

Different Examples:

Polymer	Properties	Applications
Isotactic Polypropylene (iPP)	high crystallinity, high strength	packaging, textiles, and automotive parts
Syndiotactic Polypropylene (sPP)	toughness and clarity	packaging films & automotive components
Atactic Polypropylene (aPP)	amorphous nature	adhesives, sealants, and coatings

Polymer	Properties	Applications
Isotactic Polystyrene (iPS)	high rigidity, high strength	-
Syndiotactic Polystyrene (sPS)	high heat & chemical resistance	electronics and automotive parts
Atactic Polystyrene (aPS)	amorphous nature	disposable consumer goods & packaging materials

Polymer	Properties	Applications
Isotactic Poly(methyl methacrylate) (iPMMA)	improved mechanical properties	optical applications
Syndiotactic Poly(methyl methacrylate) (sPMMA)	optical clarity	used in lenses and optical devices
Atactic Poly(methyl methacrylate) (aPMMA)	amorphous nature	Used in applications requiring flexibility

Q.4: Write down three examples of Thermoplastic and Thermosetting polymers.

Ans: Thermoplastics and thermosetting polymers are two different types of polymers showing different behavior upon heating and have different structures.

Thermoplastic Polymers: Thermoplastics are polymers that soften and become moldable upon heating and solidify upon cooling. This process is reversible, allowing them to be reshaped multiple times without significant chemical changes.

Examples: i) Polyethylene (PE): It is a simple polymer consisting of repeating ethylene units $(-CH_2-CH_2-)_n$. It is a lightweight and flexible and have variable densities. It is resistant to many acids, bases, and organic solvents.

Applications: Low-Density Polyethylene (LDPE): Used in plastic bags, packaging films, and squeeze bottles.

High-Density Polyethylene (HDPE): Used in milk jugs, detergent bottles, and water pipes.



ii) Polyvinyl Chloride (PVC): It is made from repeating units of vinyl chloride $(-CH_2-CHCl-)_n$. It is durable and Strong having good mechanical strength and toughness. It is resistant to water and many chemicals. It can be made rigid or flexible by adding plasticizers.

Applications: Rigid PVC is used in construction for pipes, window frames, and siding. Flexible PVC is used in electrical cable insulation, flooring, and medical tubing.



iii) Polystyrene (PS): It consists of repeating styrene units $(-\text{CH}_2-\text{CH}(\text{C}_6\text{H}_5)-)_n$. It is transparent with low density and easy to shape.

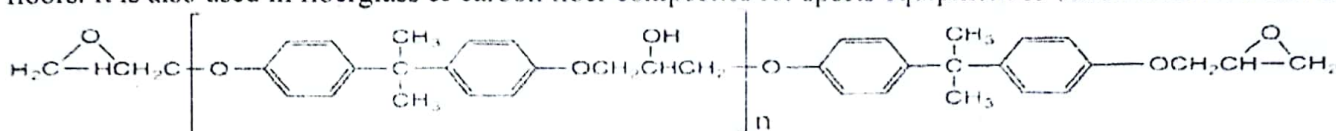
Applications: It is used in disposable cutlery, CD cases, and laboratory ware. Expanded Polystyrene (EPS) is used in packaging, insulation, and foam cups.



Thermosetting Polymers: Thermosetting polymers when heated undergo a chemical change, forming a cross-linked structure that cannot be remolded or reshaped. Once cured, these polymers become rigid and retain their shape even at high temperatures. They don't melt upon heating instead decomposed at very high temperature.

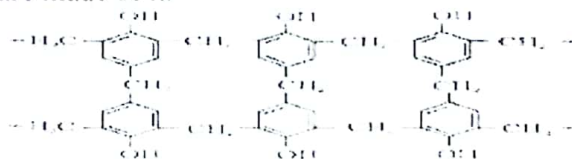
Examples: i) Epoxy Resin: Epoxy resins are formed from the reaction of epichlorohydrin and bisphenol A, resulting in a three-dimensional cross-linked network. These have excellent mechanical properties and strong adhesion. These are resistant to many chemicals, including acids and bases. These have good insulating properties.

Applications: It is used in adhesives for automotive, aerospace industries, in protective coatings for metals & floors. It is also used in fiberglass & carbon fiber composites for sports equipment & construction materials.



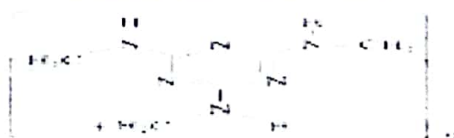
ii) Phenolic Resin (Bakelite): Phenolic resins are made from the reaction of phenol with formaldehyde, forming a highly cross-linked network. It can withstand high temperatures without deformation. It has excellent insulating properties and flame-retardant due to its aromatic structure.

Applications: It is used in circuit boards, switches, and electrical insulators. The handles, knobs of cookware, brake pads and automotive parts are made of it.



iii) Melamine Formaldehyde Resin: It is formed by the polymerization of melamine with formaldehyde, resulting in a thermosetting network. It has high hardness and surface durability, resistant to scratches, stains, and heat. It gives smooth, glossy finish to article made of it.

Applications: It is used in decorative laminates for furniture, employed in adhesives for woodworking and laminating. The melamine dinnerware and kitchen utensils are made of it.



Q.5: What is 'Living' polymerization?

Ans: "A chain polymerization from which chain transfer and chain termination are absent".

Living polymerization is a form of chain growth polymerization where the ability of a growing polymer chain to terminate has been removed. Chain termination reactions are absent, and the rate of chain initiation is also much larger than the rate of chain propagation. The result is that the polymer chains grow at a more constant rate than seen in traditional chain polymerization. Living polymerization is a popular method for synthesizing block copolymers. Since the polymer can be synthesized in stages, each stage contains a different monomer. Additional advantages are predetermined molar mass and control over end groups.

Living polymerization is desirable because it offers precision and control in macromolecular synthesis. This is important since many of the novel/useful properties of polymers result from their microstructure and molecular weight. Since molecular mass and dispersity are less controlled in non-living polymerizations, this method is more desirable for materials design.

The main living polymerization techniques are:

- i) Living anionic polymerization
- ii) Living cationic polymerization
- iii) Living ring opening metathesis polymerization
- iv) Living free radical polymerization
- v) Living chain growth polycondensation

Q.6: Differentiate between co-polymers and polyblends, give examples.

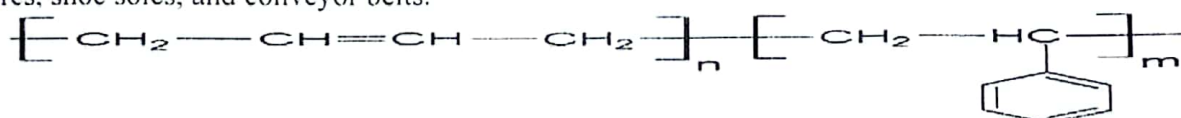
Ans: Co-polymers and polyblends are two different types of polymer materials with different characteristics and applications.

Difference Between Co-polymers and Polyblends

Property	Co-polymers	Polyblends
Definition	Polymers formed from two different monomer units chemically bonded together in a specific arrangement.	It is a physical mixture of two or more different polymers blended without chemical bonding & each component retain its original properties to some extent.
Composition	Chemical combination of different monomers.	Physical combination of existing polymers.
Chemical Bonding	Monomers are chemically bonded, resulting in a single macromolecular chain.	No chemical bonds between different polymer components; simply mixed.
Types	Block, random, alternating, graft co-polymers.	Homogeneous blends, heterogeneous blends, compatibilized blends.
Properties	Properties depend on monomer composition and arrangement.	Properties depend on the physical properties of constituent polymers.
Synthesis Method	Polymerization of two monomers.	Mechanical blending, extrusion, or other physical mixing techniques.
Structural Control	Precise control over polymer structure & properties.	Control limited to the proportion and compatibility of polymers.
Examples	Styrene-butadiene rubber (SBR), Nylon 6,6, Acrylonitrile Butadiene Styrene (ABS).	Polyethylene/Polypropylene (PE/PP) blend, Polystyrene/Poly(methyl methacrylate) (PS/PMMA) blend.
Applications	Tires, automotive parts, clothing fibers, adhesives.	Packaging, construction materials, consumer goods.

Examples of co-polymers:

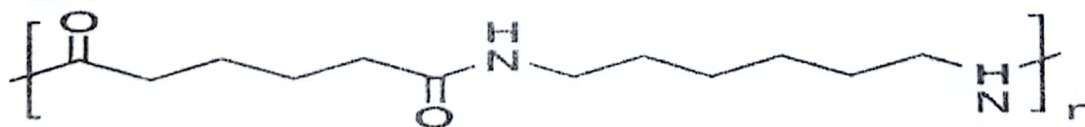
Styrene-Butadiene Rubber (SBR): It is a block co-polymer with good elasticity & abrasion resistance. It is used in car tires, shoe soles, and conveyor belts.



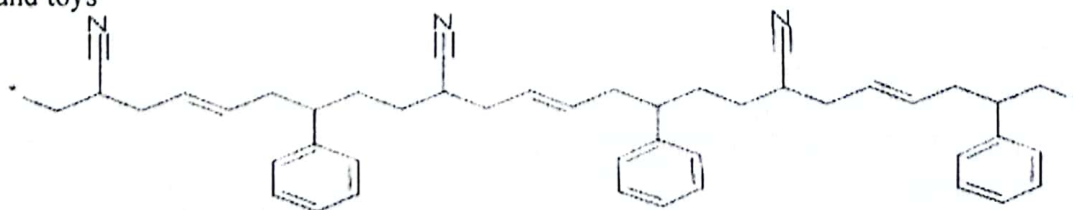
Ethylene Propylene Rubber (EPR): It is random co-polymer with excellent thermal and chemical resistance. It is used in automotive weather-stripping and hoses.



Nylon 6,6: It is an alternating co-polymer with high strength and thermal stability. It is used in textiles, carpets, and industrial fibers

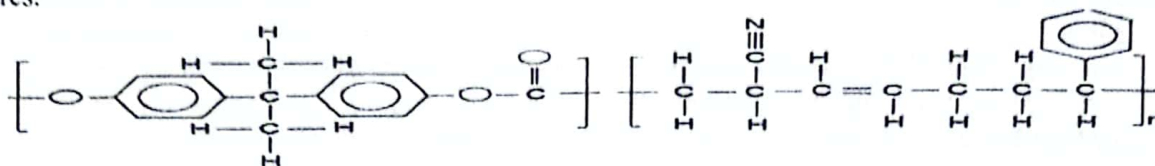


Acrylonitrile Butadiene Styrene (ABS): It is a graft co-polymer with polybutadiene backbone grafted with styrene-acrylonitrile chains. It has high impact resistance and toughness. It is used in automotive parts, electronics, and toys

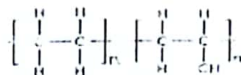


Examples of Polyblends:

Polycarbonate/ABS (PC/ABS) Blend: it is a homogeneous blend and used in automotive parts and electronic enclosures.



Polyethylene/Polypropylene (PE/PP) Blend: It is a heterogeneous blend with enhanced mechanical properties. It is used in packaging and consumer goods



Polystyrene/Poly(methyl methacrylate) (PS/PMMA) Blend: it is a compatibilized blend with improved mechanical properties and transparency. It is used in optical devices and display technologies.



Q.7: Differentiate between 'Atactic' & 'isotactic' polymers.

Ans: Atactic and isotactic polymers are two types of stereoisomers having spatial arrangement of their substituent groups along the polymer chain

comparison of atactic and isotactic polymers

Property	Atactic Polymers	Isotactic Polymers
Definition	Polymers with substituent groups are randomly oriented along the polymer chain.	Polymers with substituent groups are arranged on the same side of the polymer chain in a regular, repeating pattern.
Structure		
Stereochemistry	Random configuration of chiral centers with no consistent pattern.	Consistent stereochemistry with identical configurations at all chiral centers.
Crystallinity	Amorphous with low crystallinity due to the random arrangement.	Highly crystalline due to regular and uniform arrangement with tight packing.
Melting Point	Lower melting points due to amorphous structure and lack of crystallinity.	Higher melting points due to crystalline structure and stronger intermolecular forces.
Mechanical Properties	softer and more flexible due to amorphous nature.	Rigid and strong, with enhanced mechanical properties such as tensile strength and stiffness.
Optical Properties	Often translucent or opaque, as the irregular structure scatters light.	More likely to be clear and transparent due to uniform structure.
Solubility	more soluble in solvents due to the lack of crystalline regions.	Less soluble due to tightly packed crystalline structure, which resists solvent penetration.
Tacticity Configuration	Randomly alternating R and S configurations, leading to no overall tacticity.	Consistent R or S configuration throughout the polymer chain, resulting in uniform tacticity.
Production Method	Typically produced via free radical polymerization, leading to random configurations.	Usually synthesized using specific catalysts (e.g., Ziegler-Natta or metallocene catalysts) to control stereochemistry.
Example	Atactic Polypropylene (aPP): Used in adhesives, sealants, and roofing due to its flexibility and low cost.	Isotactic Polypropylene (iPP): Used in automotive parts, textiles, and packaging for its strength and durability.
Thermal Properties	Lower thermal stability and resistance to deformation under heat.	Higher thermal stability and resistance to heat due to crystalline regions.
Density	Lower density due to irregular packing and structure.	Higher density due to orderly packing and crystalline arrangement.
Applications	Primarily used in products where flexibility, low cost, and easy processing are desired.	Ideal for applications requiring rigidity, high strength, and thermal resistance.

Q.8: What are the main applications of polystyrene?

Ans: Polystyrene is a thermoplastic polymer and have applications in various industries. Main Applications of Polystyrene are:

i) Packaging:

Food Packaging: Cups, plates, cutlery, and food containers e.g. foam cups for hot beverages, clamshell containers for takeout meals, and trays for meat products.

Protective Packaging: Foam cushioning for electronics, appliances, and delicate items.

Insulated Packaging: Ice chests, coolers, and thermal containers e.g. Coolers for transporting vaccines, iceboxes for camping and picnics

ii) Consumer Goods:

Household Items: Utensils i.e. plastic cutlery for outdoor events, storage bins for organizing household items, disposable razors, cosmetic containers, and storage boxes

Electronics Housings: Enclosures for remote controls, battery cases, casings for electronic devices like calculators and portable speakers.

Stationery Products: Transparent rulers for schools, pen casings, and desktop organizers.

iii) Construction:

Insulation Boards: EPS and XPS foam boards for insulation walls, roofs, and foundations in residential and commercial buildings.

Lightweight Fill Material: Fill material for roads, embankments, and landscaping

iv) Electrical and Electronics:

Insulation Components: Insulating components in switches, sockets, and circuit boards.

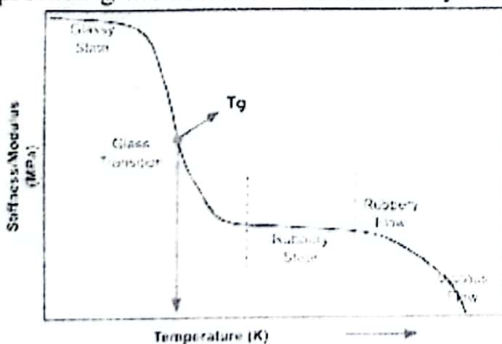
Electronics Enclosures: Casings for small electronics like clocks, radios, and battery packs.

v) Medical and Healthcare: Sterile trays, pill bottles, and medical device packaging i.e. packaging for syringes, vials, and diagnostic equipment. Petri dishes for bacterial culture, test tubes, and disposable pipette tips, disposable syringes, surgical instruments, and specimen containers.

vi) Automotive: Dashboard panels, Instrument panels, door panels, and decorative trims, knobs, bumpers and exterior trims.

Q.9: What do you understand about glass transition temperature of polymers?

Ans: The glass transition temperature (T_g) is a temperature range over which a polymer changes from a hard, glassy material to a softer, more rubbery state. The glass transition temperature (T_g) is the temperature at which an amorphous polymer (or the amorphous regions of a semi-crystalline polymer) changes from a brittle, glassy state to a flexible, rubbery state. It is not a single temperature point but rather a range over which this transition occurs. Below T_g a polymer exhibit a hard and glassy behavior. The molecular chains are in a frozen state, with very little molecular motion. The material is rigid and can fracture easily under stress. Above T_g a polymer become soft and rubbery. The increased thermal energy allows the molecular chains to move more freely, providing the material with flexibility and elasticity.



Polymer	(T_g) [$^{\circ}\text{C}$]
Polystyrene (PS)	95°C
Poly(methyl methacrylate) (PMMA)	105°C
Polyvinyl Chloride (PVC)	82°C
Polyethylene Terephthalate (PET)	70°C
Polycarbonate (PC)	150°C
Nylon 6,6	50°C
Polypropylene (PP)	-20°C
Polyurethane (PU)	-40°C to 0°C

Factors Affecting Glass Transition Temperature (T_g):

- Molecular Weight:** Higher molecular weight polymers generally have higher T_g due to increased chain entanglement and reduced mobility.
- Plasticizers:** Adding plasticizers lowers T_g by reducing intermolecular forces and increasing chain mobility. This is commonly used to make PVC more flexible.
- Copolymers:** Copolymerizing with different monomers can modify T_g . For example, adding a flexible monomer can lower T_g .
- Crosslinking:** Increased crosslinking restricts chain movement, leading to higher T_g as in thermosetting polymers.
- Side Groups and Branching:** Bulky or polar side groups hinder chain mobility, increasing T_g . Branching can either raise or lower T_g depending on the structure.

Q.10: How is PVC produced? Give its uses.

Ans: PVC (polyvinyl chloride) is produced by the polymerization of vinyl chloride. The production process generally involves two main steps:

i) Production of Vinyl Chloride Monomer (VCM):

The parent compound, vinyl chloride may be prepared by any of the following methods:

- $\text{HC}\equiv\text{CH} + \text{HCl} \xrightarrow{\text{HgCl}_2/100-200^\circ\text{C}} \text{ClCH}=\text{CH}_2$
Acetylene Vinyl Chloride
- $\text{CH}_2=\text{CH}_2 + \text{Cl}_2 \rightarrow \text{ClCH}_2\text{CH}_2\text{Cl} \xrightarrow{\text{alkali } 300-600^\circ\text{C}} \text{ClCH}=\text{CH}_2 + \text{HCl}$
Ethylene ethylene dichloride Vinyl Chloride
- $\text{CH}_2=\text{CH}_2 + \frac{1}{2}\text{O}_2 + \text{HCl} \xrightarrow{\text{CuCl}_2/500^\circ\text{F}} \text{ClCH}=\text{CH}_2 + \text{H}_2\text{O}$
Ethylene Vinyl Chloride

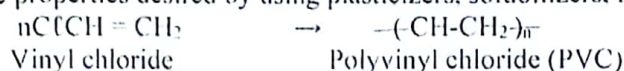
ii) Polymerization of VCM to PVC: The polymerization of VCM to form PVC can be carried out using different methods:

i) Suspension Polymerization

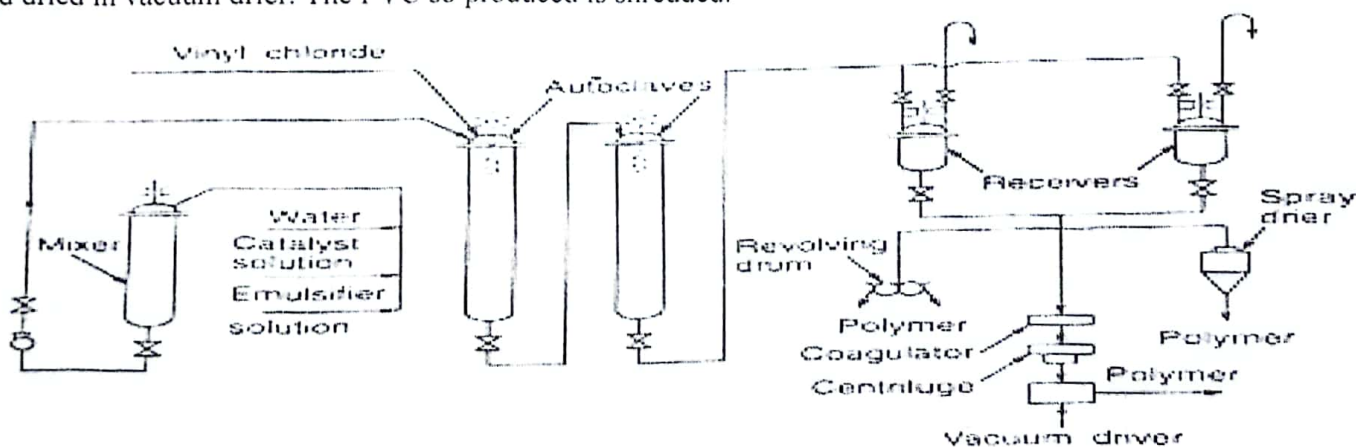
ii) Emulsion Polymerization

iii) Bulk Polymerization

Emulsion Polymerization: A mixture of liquid vinyl chloride (100 parts), water (100 parts), Sodium per sulphate, potassium per sulfate, catalyst (1 part) and an emulsifier, such as sodium lauryl sulphate (1.5 part) is vigorously shaken in the form of tiny globules and then introduced into an autoclave maintained at 40-45 °C. The reaction is completed in about 72 hours. The polymer particles are then recovered by either spray drying or by coagulation by an acid solution. The PVC polymer may be molded according to the properties desired by using plasticizers, solubilizers, lubricants and fillers.



Water and solution of an emulsifier and initiator are fed from a mixer into a nickel-plated autoclave provided with a jacket and an agitator. Next vinyl chloride is introduced under a rarefaction. At the beginning of the process, the temperature is maintained at about 45 °C by passing warm water through the jacket. The heat of the exothermic reaction is removed by brine with a temperature of -20°C circulating in the jacket and the mixer. The pressure in the autoclave is about 0.5 MPa and duration of the process is 40-60 hours. The polymer emulsion from the autoclave is transferred to another autoclave and continuously gathered in receivers containing soda as stabilizer. The PVC can be separated from the emulsion by spraying on revolving drums, in spray driers, or in coagulator. $\text{Al}_2(\text{SO}_4)_3$ is used as a coagulant. Next the polymer is washed with water, squeezed out in centrifuge, and dried in vacuum drier. The PVC so produced is shredded.



Production of PVC by continuous emulsion method.

Applications of Polyvinyl Chloride (PVC)

- It is also used as electrical insulation material for cable application especially sheathing (cover over the insulator of electrical wire) material.
- The unplasticized PVC is used in chemical plants and equipment, storage, tank, building items like pipes, sheets, door, frame, containers, roof sheet, floor tiles, wall lining, etc. semirigid and flexible application includes packaging, tubes, flexible pipes, hoses, sheets, films, footwear, belting, cables, baby pants, toys, etc.

Q.11: Differentiate between co-polymers and terpolymers.

Ans: Co-polymers and terpolymers are both types of polymers made from more than one kind of monomer, but they differ in the number of monomers used.

Co-polymers: Co-polymers are polymers that are made from two different types of monomers.

Types:

Random Co-polymers: Monomers are randomly distributed along the polymer chain. -A-B-A-A-B-B-A-B-

Alternating Co-polymers: Monomers alternate in a regular pattern. -A-B-A-B-A-B-

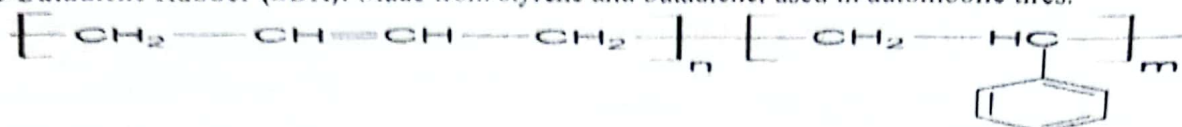
Block Co-polymers: Large blocks of each type of monomer are grouped together. -A-A-A-B-B-B-A-A-

Graft Co-polymers: Chains of one type of monomer are grafted onto the backbone of another type.

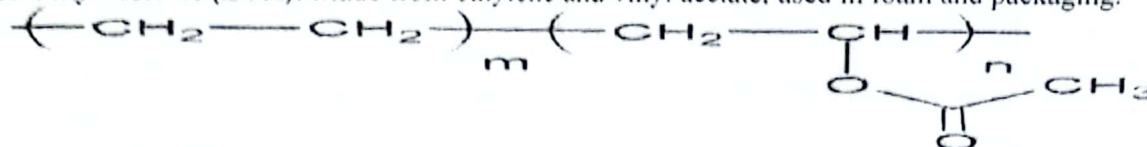
-A-A-A-A-B-B-B-

Examples:

Styrene-Butadiene Rubber (SBR): Made from styrene and butadiene, used in automobile tires.



Ethylene-Vinyl Acetate (EVA): Made from ethylene and vinyl acetate, used in foam and packaging.



Terpolymers: Terpolymers are polymers that are made from three different types of monomers.

Types:

Random Terpolymers: Three monomers are randomly distributed along the polymer chain.

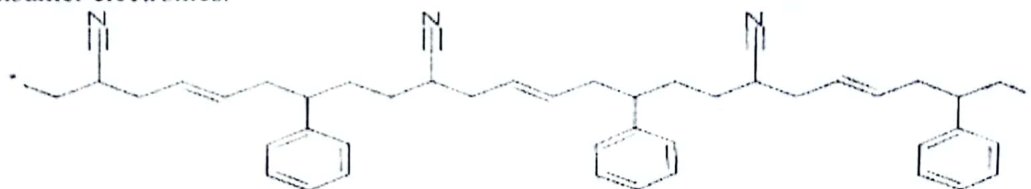
-A-B-C-A-C-B-A-B-A-B-C-A-C-B-A-B-A-B-C-A-C-B-A-B-

Block Terpolymers: Blocks of three different monomers are grouped together. -A-A-B-B-C-C-

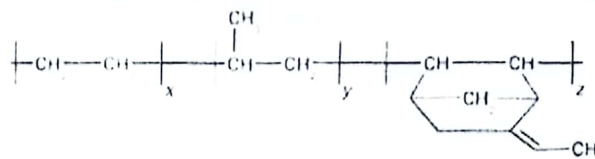
Graft Terpolymers: Chains of two types of monomers are grafted onto the backbone of the third monomer type.

Examples:

Acrylonitrile Butadiene Styrene (ABS): Made from acrylonitrile, butadiene, and styrene, used in automotive parts and consumer electronics.



Ethylene-Propylene-Diene Monomer (EPDM): Made from ethylene, propylene, and a diene component, used in weather-sealing and roofing membranes.



Q.12: Differentiate between Thermoplastic and Thermosetting polymers.

Ans: Difference between thermosetting plastic and thermoplastic

Sr. #	Thermoplastic plastic	Thermosetting plastic
1	They can be softened on heating & hardened on cooling reversibly. Hence their hardness is temporary and can be changed by cooling or heating (decreasing or increasing the temperature), so they are called thermoplastic polymers.	They are hardened during molding or heating and once solidified, they cannot be softened again and hence called permanent thermosetting polymers.
2	They can be formed by addition polymerization only.	They can be formed by condensation polymerization.
3	They are generally long chain linear polymers.	They have three-dimensional crosslinked network structure joined by strong covalent bonds.
4	On heating, they soften readily because secondary forces between the chain can be easily broken.	They don't soften on heating, because cross links and strong covalent bonds retain their strength on heating. They can, however, be charred on strong heating.
5	They can be softened, reshaped and even reused by reheating to a suitable temperature.	They cannot be reshaped and reused because they do not soften on heating and retain their shape and structure.
6	They are generally weak, soft and less brittle	They are usually hard, strong and more brittle
7	They can be reclaimed from wastes.	They cannot be reclaimed from wastes.
8	They can dissolve in some organic solvents.	They are insoluble almost in all organic solvents because of the presence of cross linking and strong covalent bonds.
9	The thermoplastics are cellulose nitrate, cellulose acetate butyrate, ethyl cellulose, polyacrylates, polyvinyl resins, polystyrene resins, polyamides (nylon), polyethers, polypropylene, polyethylene etc.	The thermosetting are phenol resins (phenol formaldehyde, phenol furfural etc.), amino resins (Urea formaldehyde, melamine formaldehyde etc.), polyesters (polycarbonates, epoxides etc.)

Q.13: Name two important techniques of polymer fabrication.

Ans: **Polymer fabrications:** Polymer fabrication means various processes used to create finished products from polymer materials.

- Fabrication includes techniques and methods to shape, form, and assemble polymers into desired forms, structures, and objects.
- In polymer fabrication, raw polymer materials such as granules, pellets, powders, or resins, is transformed into usable items having specific functions.

Important Techniques of Polymer Fabrication:

- Injection Molding:** Injection molding is a manufacturing process used to produce parts by injecting molten material into a mold. It is one of the most common methods for producing plastic products with complex shapes and fine details.
- Extrusion:** Extrusion is a continuous manufacturing process in which polymer material is forced through a shaped die to produce long objects of uniform cross-section. This technique is ideal for creating items like pipes, tubing, and sheets.
- Blow Molding:** It is a technique to create hollow objects like bottles and containers by inflating a molten polymer tube inside a mold.
- Thermoforming:** It is a technique in which a plastic sheet is heated to make pliable (soft) and then forming it over a mold and trimming to create the final shape.

Q.14: Explain the difference between homopolymers and co-polymers.

Ans: Homopolymers and copolymers are two distinct types of polymers, distinguished by the arrangement and types of monomers that constitute their structure.

Sr.#	Homopolymer	Copolymer
1	Polymers made up of only one type of monomer unit repeated throughout the entire chain.	Polymers made from two different types of monomers combined in the same polymer chain.
2	They have a uniform structure due to a single type of monomer.	Copolymers may have regular, random, alternating and block structures.
3	The homopolymers have predictable and consistent physical, chemical, and mechanical properties which are determined by the single monomer.	Copolymers have diversity in properties and functionalities. The types and ratios of monomers in copolymers give specific characteristics as compared to homopolymers
4	Examples: i) Polyethylene (PE) [monomer Ethylene (C_2H_4)] used in Plastic bags, bottles, and piping ii) Polystyrene (PS) [monomer Styrene (C_8H_8)] used in Disposable cutlery, CD cases, and foam insulation iii) Polyvinyl Chloride (PVC) [monomer Vinyl chloride (C_2H_3Cl)] used in Pipes, electrical cable insulation, and clothing iv) Polytetrafluoroethylene (PTFE) [monomer Tetrafluoroethylene (C_2F_4)] used in Non-stick coatings for cookware, lubricants, and gaskets. v) Polymethyl Methacrylate (PMMA) [monomer Methyl methacrylate ($C_5H_8O_2$)] used in Acrylic glass, lenses, and screens. vi) Polypropylene (PP) [Propylene (C_3H_6)] used in Packaging, textiles, and automotive parts.	Examples: i) Nylon-6,6 [monomers Hexamethylene diamine and adipic acid] used in Textiles, carpets, and industrial components ii) Styrene-Butadiene Rubber (SBR) [monomers Styrene and butadiene] used in Automobile tires, footwear, and adhesives iii) Ethylene-Vinyl Acetate (EVA) [monomers Ethylene and vinyl acetate] used in Foam in sports equipment, packaging, and footwear. iv) Polyethylene Terephthalate (PET) [monomers Ethylene glycol and terephthalic acid] used in Beverage bottles, food packaging, and synthetic fibers.

- What do you understand about coordination polymerization, briefly explain its importance?
- What are epoxy resins? Give their important applications.
- What is Free radical polymerization?
- Write two major differences between chain growth and step growth polymerization methods.
- What type of catalysts are used in anionic polymerization, give examples?
- What is Fabrication of Plastics?